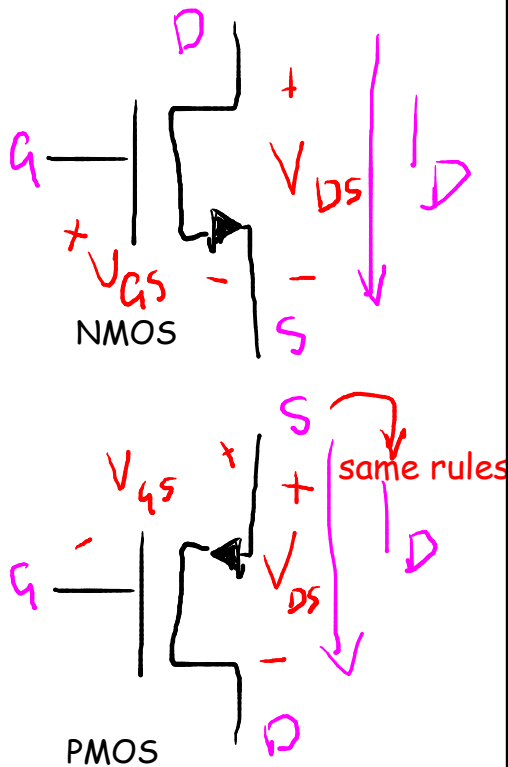


steps for MOSFET analysis questions:

1. DC analysis



TRIODE

$$V_{GS} > V_{TH}$$

$$I_D = \frac{1}{2} K_n [2V_{ov} V_{DS} - V_{DS}^2]$$

where:

$$V_{ov} = V_{GS} - V_{TH}$$

SATURATION

$$V_{GS} > V_{TH} \text{ AND } V_{DS} > V_{ov}$$

$$I_D = \frac{1}{2} K_n V_{ov}^2 (1 + \lambda V_{DS})$$

CLM (lambda is zero when neglected)

$$K_n = \mu_n C_{ox} \frac{W}{L}$$

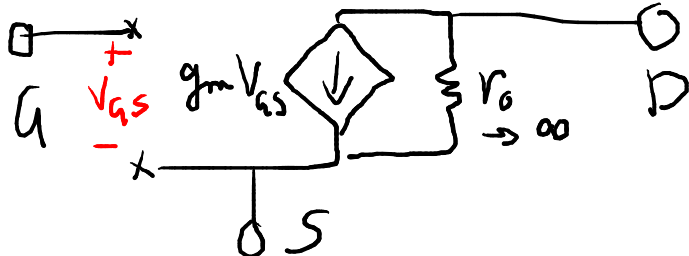
2. gm calculations

$$g_m = \sqrt{2K_n I_D} = 2 \frac{I_D}{V_{ov}} = K_n V_{ov}$$

$$r_o = \frac{1}{\lambda I_D}$$

3. AC analysis

A. small signal model



B- GmRout method

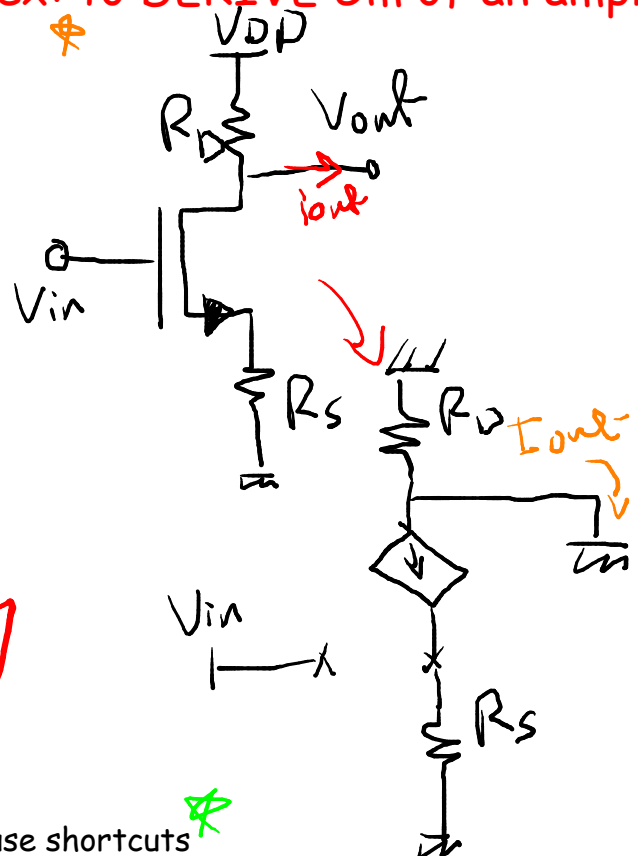
$$A_v = G_m R_{out}$$

volt gain

$$G_m = \frac{i_{out}}{v_{in}} \text{ when } i_{out} \text{ is grounded}$$

$$R_{out} = \frac{v_{out}}{i_{out}} \text{ or use shortcuts}$$

ex: to DERIVE Gm of an amplifier



to memorize Gm

Summary of Basic Topologies

	CS	CG	CD (SF)
	Voltage & current amplifier	Voltage amplifier Current buffer	Voltage buffer Current amplifier
Rin	∞	$R_S \parallel \frac{1}{g_m + \frac{1}{r_o}} \left(1 + \frac{R_D}{r_o}\right)$	∞
Rout	$R_D \parallel r_o [1 + (g_m + \frac{1}{r_o}) R_S]$	$R_D \parallel r_o$	$R_S \parallel \frac{1}{g_m + \frac{1}{r_o}} \left(1 + \frac{R_D}{r_o}\right)$
Gm	$\frac{-g_m}{1 + (g_m + \frac{1}{r_o}) R_S}$	$g_m + \frac{1}{r_o}$	$\frac{g_m}{1 + R_D/r_o}$

06: Basic Amplifier Stages

43

$X =$ مش علينا
gmb
(neglect)

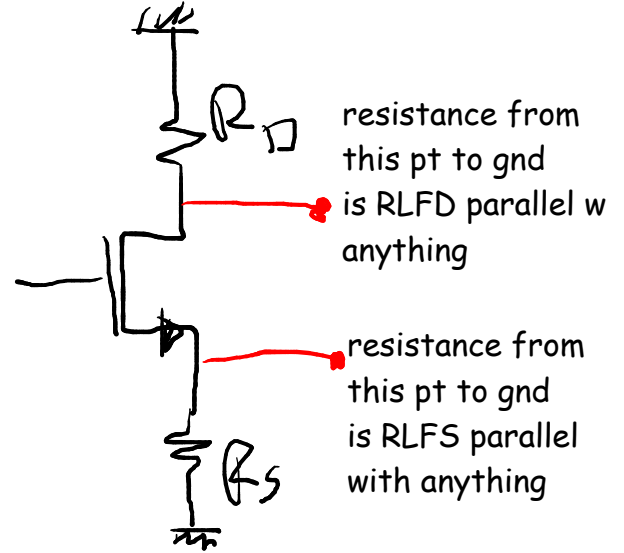
(Dr. Hesham Omran slides)

B.2- Rout shortcut (updated)

$$R_{LFD} = r_o \left[1 + \left(g_m + \frac{1}{r_o} \right) R_S \right]$$

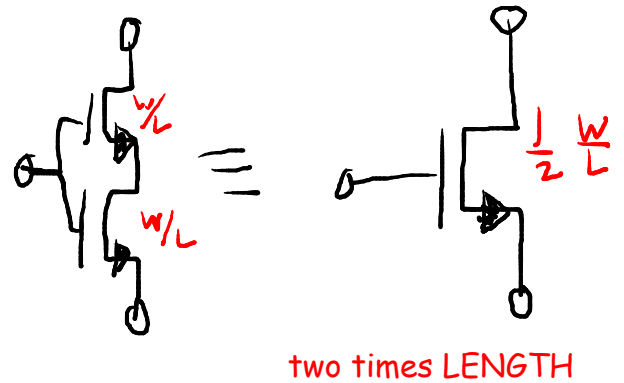
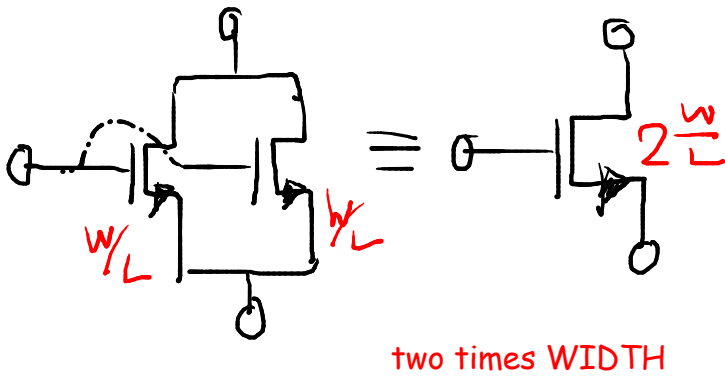
$$R_{LFS} = \frac{1}{g_m + \frac{1}{r_o}} \left[1 + \frac{R_D}{r_o} \right]$$

for

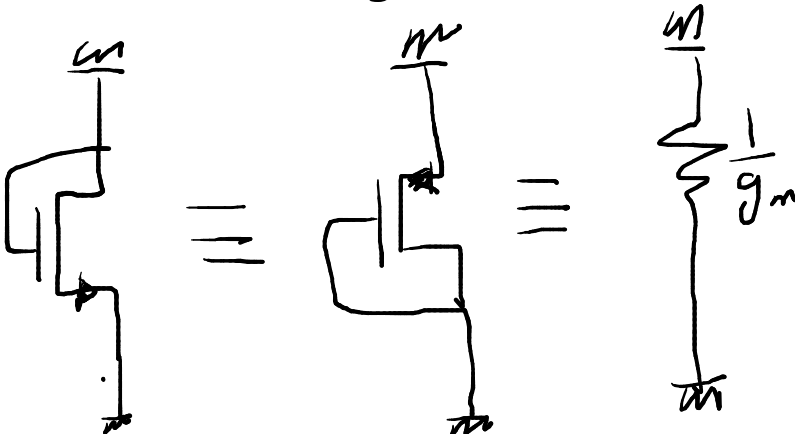


Other notes:

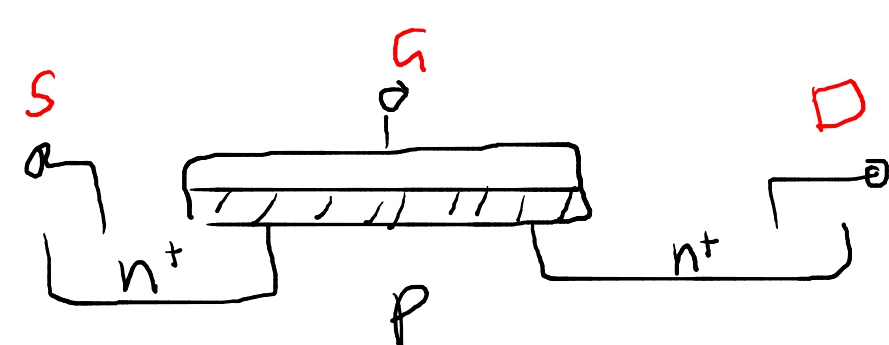
1. MOSFETs in parallel and series:



for small signal model:

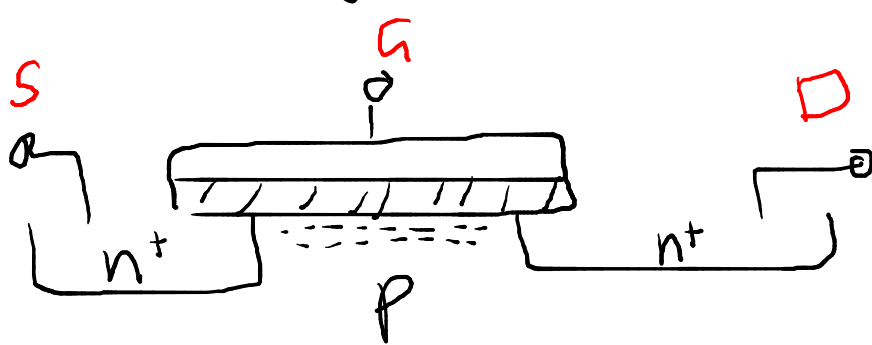


mosfet theory

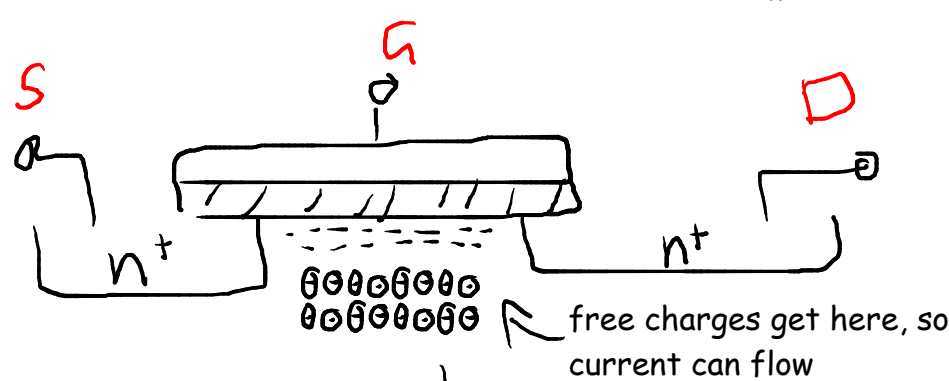


* SYMMETRIC
DEVICE

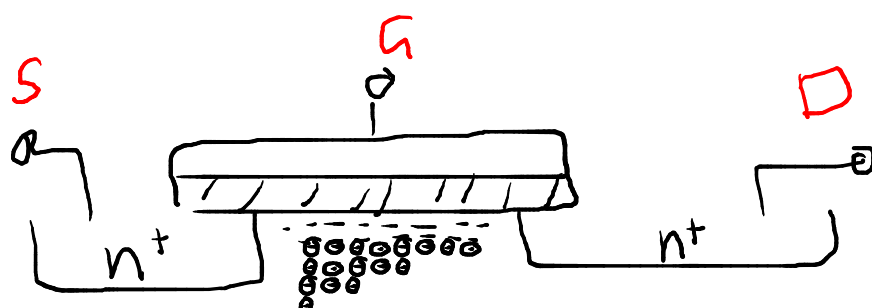
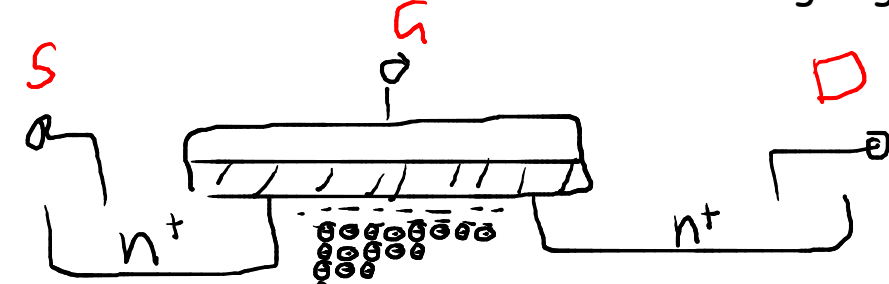
increase V_G



increase V_G even more



bias V_{DS} so charges go to S



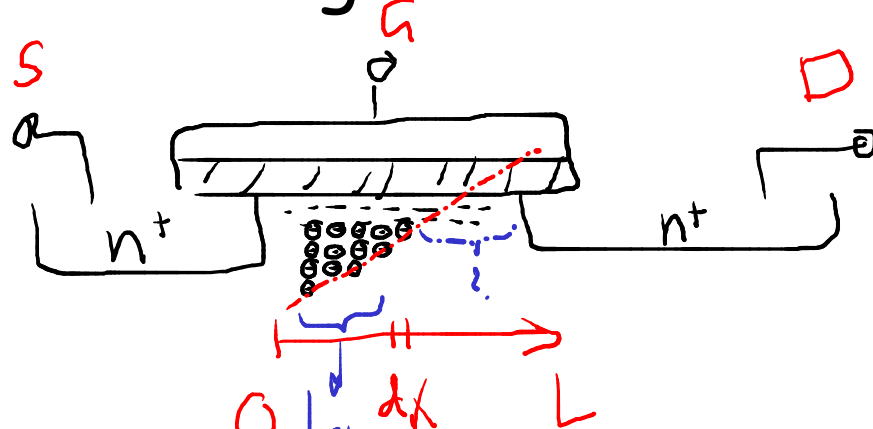
$$Q = W C_{ox} [V_{gs} - V_{th} - V(x)]$$

$$I = Q \mu_n \frac{dV}{dx}$$

$$\rightarrow \int_0^L I dx = \int_0^{V_{gs}} \mu_n C_{ox} W [(V_{gs}) - V(x)] dV$$

$$\rightarrow I = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} [2V_{ov} V_{gs} - V_{gs}^2]$$

for saturation region

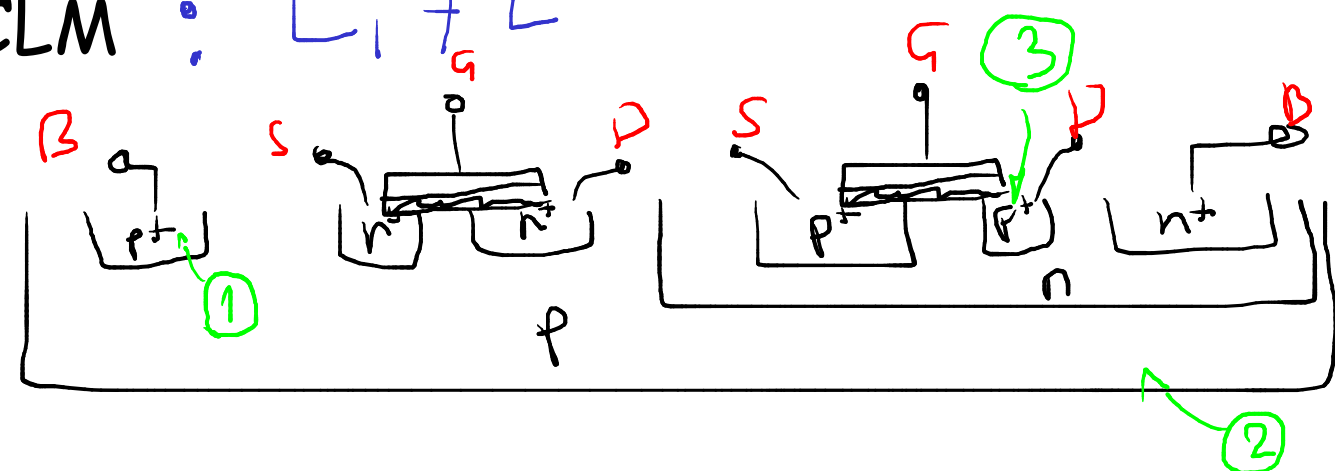


$$\int_0^L I dx = \int_0^{V_{gs} - V_{th}} \mu_n C_{ox} W [(V_{gs}) - V(x)] dV$$

$$\rightarrow I = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} [V_{ov}]^2$$

~~CLM~~ : $L_1 = L$

CLM : $L_1 \neq L$

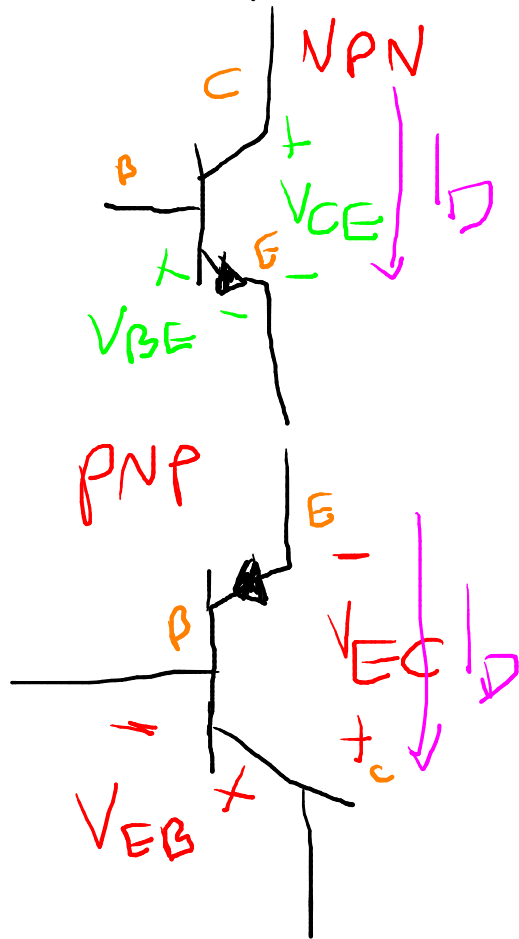


1 & 3 : they are n+ and p+ to prevent schottky effect (sudden change from doped to conductive)

2 : for silicon-heavy chips (?)

steps for BJT analysis questions

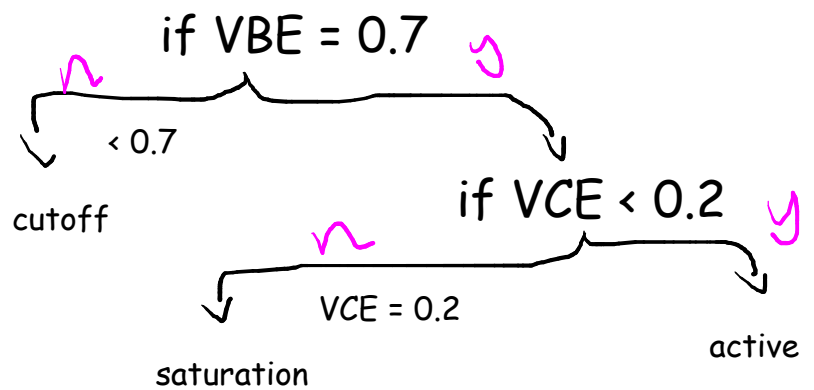
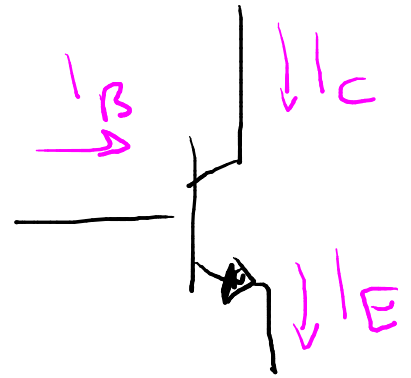
1- DC analysis



$\beta \rightarrow \text{given}$

$$I_E = \frac{\beta}{\beta + 1} I_B$$

$$I_C = \beta I_B$$



$$I_C = I_o e^{\frac{V_{BE}}{V_T}}$$

2- AC params

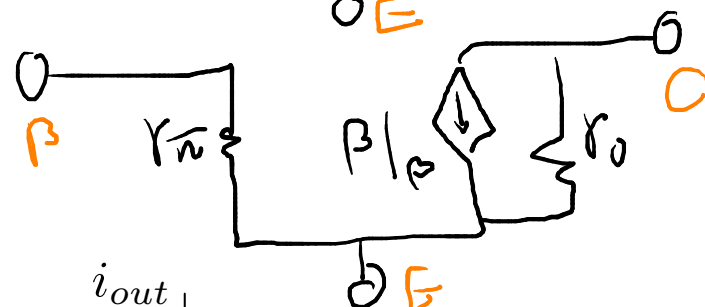
$$g_m = \frac{I_C}{V_T} \quad r_\pi = \frac{\beta}{g_m} \quad r_o = \frac{|V_A|}{I_C}$$

3- AC analysis

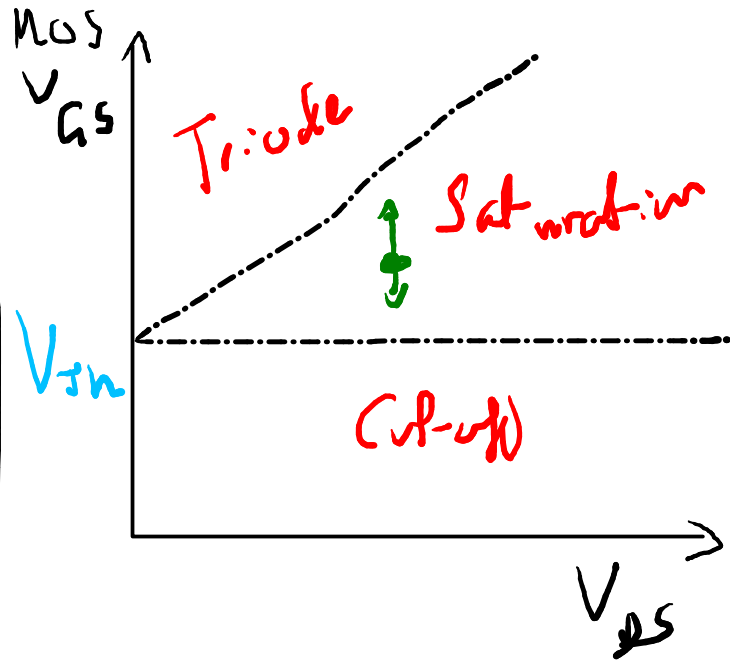
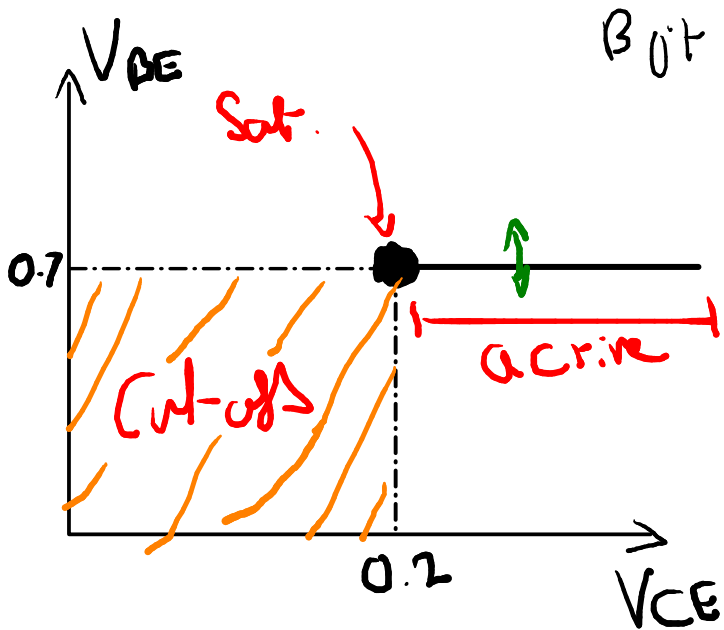
- 1- all DC blocking caps are s.c.
- 2- all volt sources are grounded
- 3- all current sources are opened
- 4- hybrid-pi model and get the five params:

$$A_v = \frac{v_{out}}{v_{in}} \quad A_i = \frac{i_{out}}{i_{in}}$$

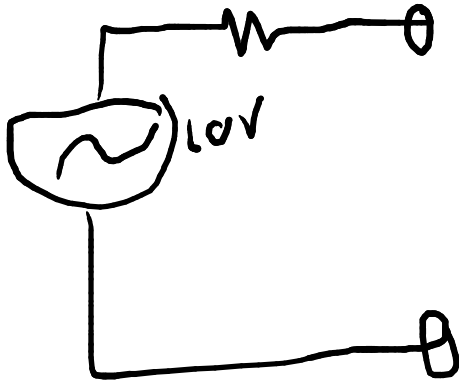
$$R_{out} = \frac{v_{out}}{i_{out}} \quad R_{in} = \frac{v_{in}}{i_{in}}$$



$$G_m = \frac{i_{out}}{v_{in}} \Big|_{I_{out} \text{ is grounded}}$$

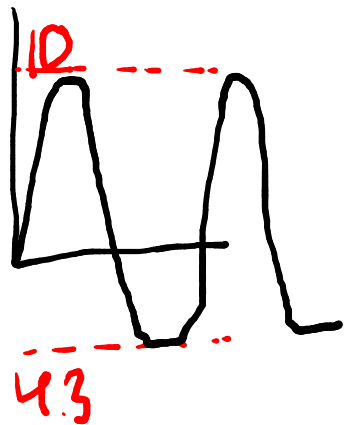
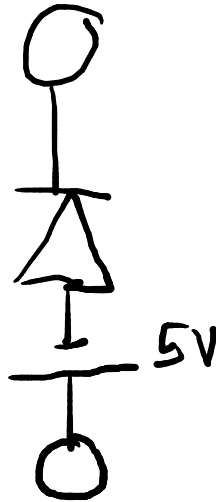
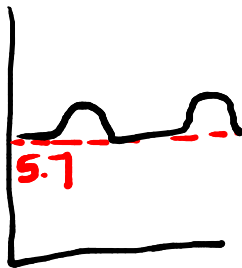
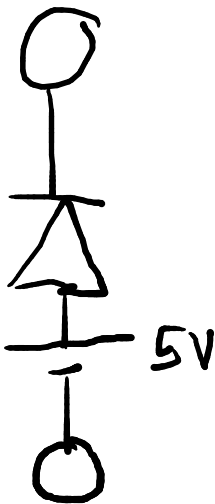
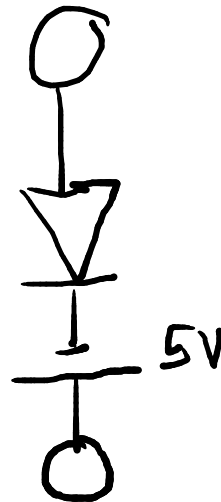
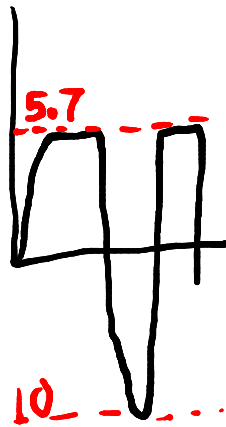


clipping circuits



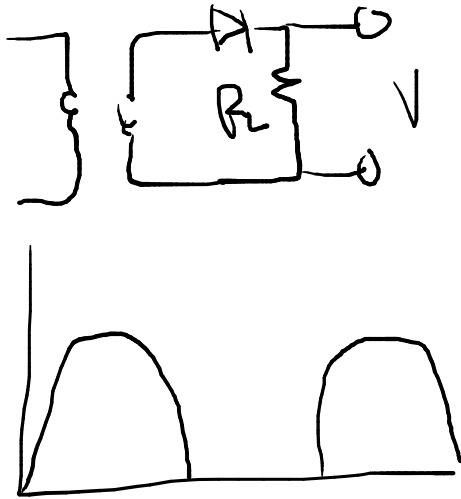
revert diode: revert direction of clipping

revert batt: revert line of clipping

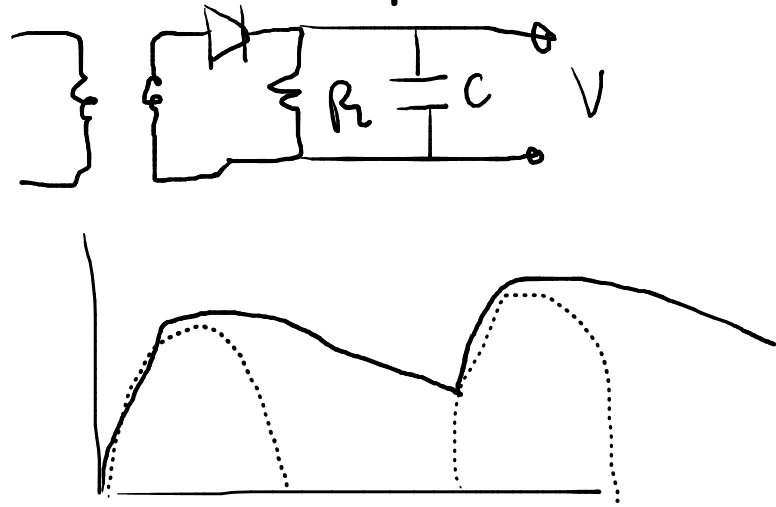


rectifiers

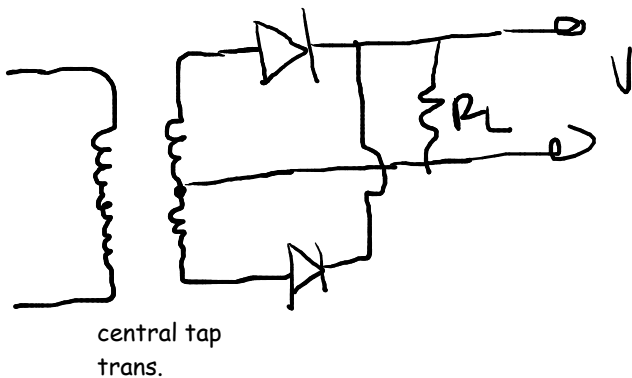
h.w.



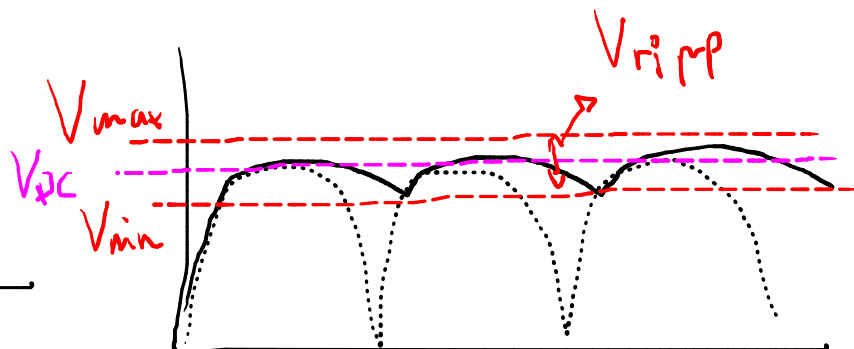
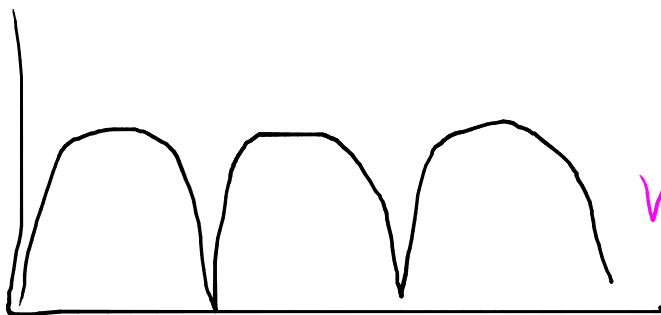
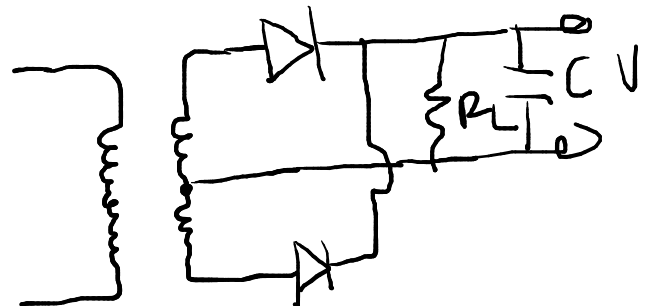
h.w. with cap



f.w.



f.w. with cap



RULES:

$$V_{\text{ripple}} = \frac{I_{\text{DC}}}{2 f C}$$

replace every $2f$ with f
for h.w. with cap

$$V_{\text{regulate}} = \frac{1}{f C R}$$

$$V_{\text{DC}} = I_{\text{DC}} R$$

$$= V_{\text{max}} - \frac{1}{2} V_{\text{ripple}}$$

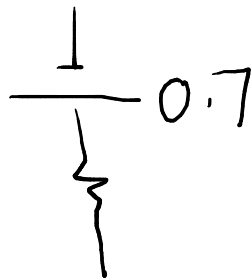
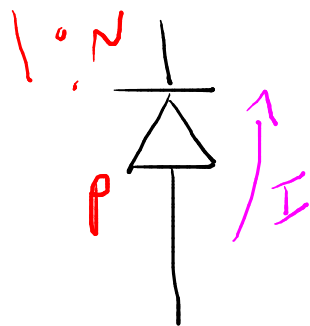
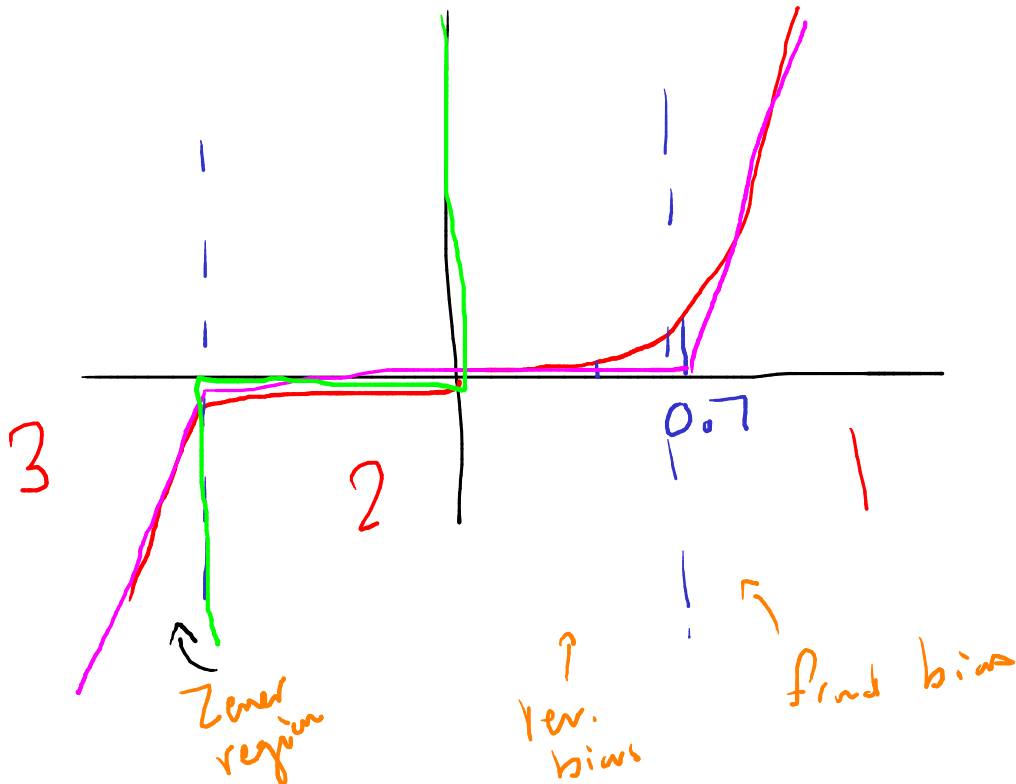
$C \uparrow \rightarrow$ higher load $\rightarrow$ lower R_L

Diode and Zener

 : actual

 : linear

 : ideal



$$I = I_{oe} e^{\frac{V}{V_T}}$$

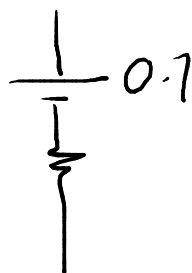
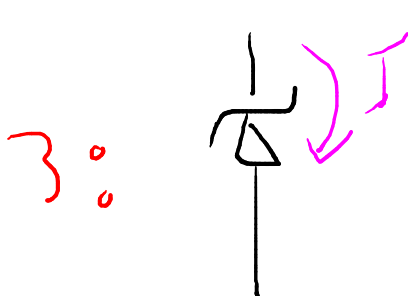
depends on area



to solve diode ct.s:

1- make assumption by inspection

2- you must verify it by looking at the sign of I



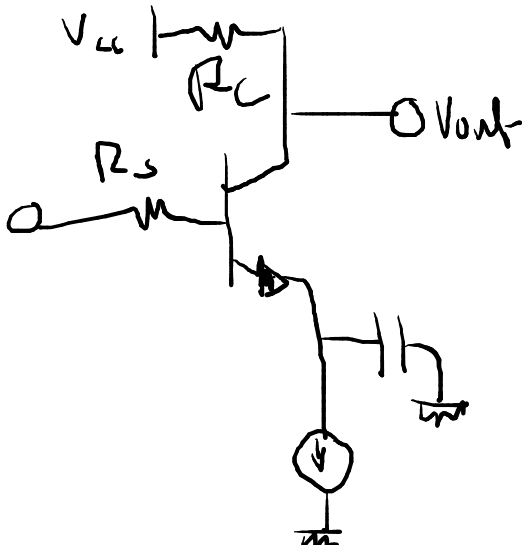
zener is used for volt regulation due to it surviving breakdown region

Amplifier types

BJT

MOSFET

1- Common Emitter



$$A_v = \frac{-\beta \times R_C \parallel R_L}{R_s + R_{in}}$$

high

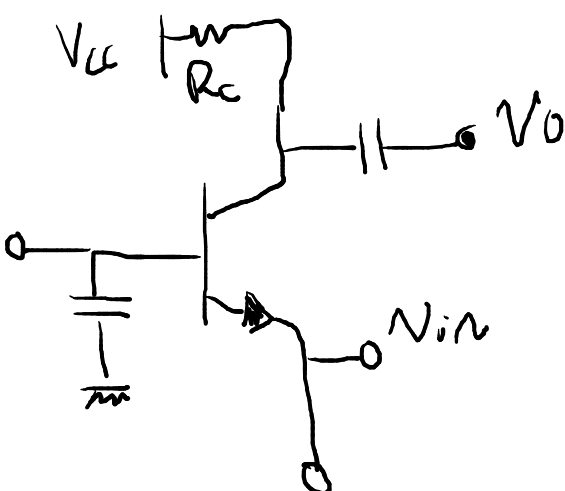
$$A_i \approx -\beta$$

$$R_{in} \approx R_{in} \quad \text{! low}$$

$$R_{out} = \frac{1}{R_C} + \frac{1}{R_L} \quad \checkmark \text{ high}$$

used in the second stage of a 3-stage amplifier due to its low inp. impedance

2- Common Base



$$A_v = \frac{\beta R_C}{r_{in} + (1 + \beta) R_s}$$

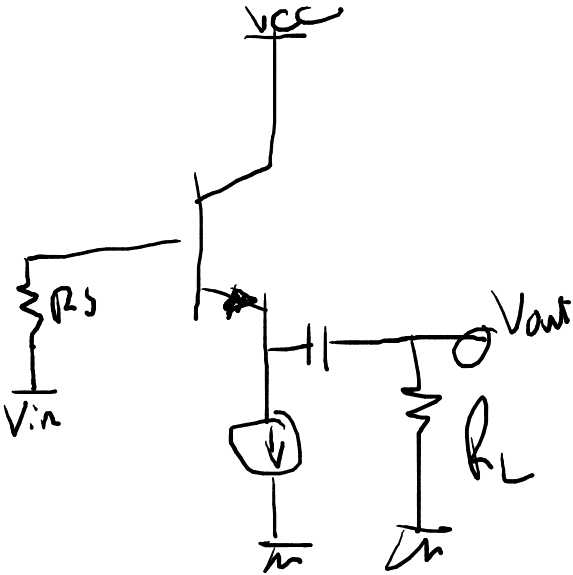
$$R_{in} = \frac{r_{in}}{\beta + 1} \quad \text{! low}$$

$$A_i = \alpha \quad [\leq 1]$$

$$R_{out} = R_C$$

used as a current buffer (high Rout)

3- Common Collector (Emitter Follower)



$$A_v = \text{calculate manually}$$

$$A_i = 1 + \beta$$

$$R_{in} = \checkmark \text{ high input imp.}$$

$$R_{out} = \checkmark \text{ low output impedance}$$

can be used at either the first stage or the third stage of a 3 stage amp due to desired impedances

high imp at input: doesnt load the circuit much

low imp at output: load doesnt affect it much

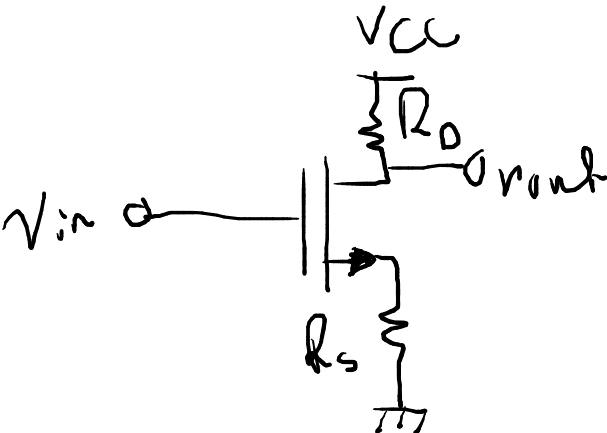
Summary of Basic Topologies

	CS	CG	CD (SF)
	Voltage & current amplifier	Voltage amplifier Current buffer	Voltage buffer Current amplifier
Rin	∞	$R_S \parallel \frac{1}{g_m + g_{mb}} \left(1 + \frac{R_D}{r_o} \right)$	∞
Rout	$R_D \parallel r_o [1 + (g_m + g_{mb}) R_S]$	$R_D \parallel r_o$	$R_S \parallel \frac{1}{g_m + g_{mb}} \left(1 + \frac{R_D}{r_o} \right)$
Gm	$\frac{-g_m}{1 + (g_m + g_{mb}) R_S}$	$g_m + g_{mb}$	$\frac{g_m}{1 + R_D / r_o}$

06: Basic Amplifier Stages

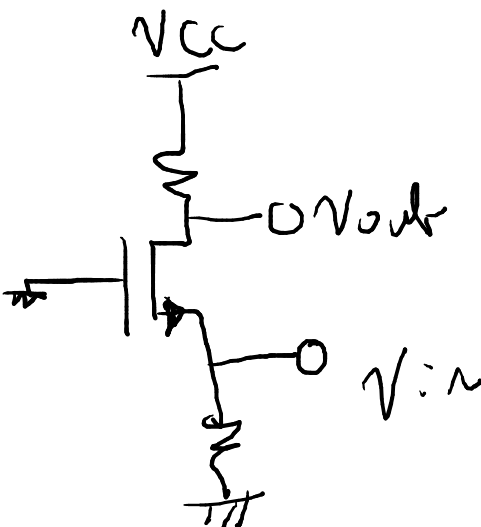
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1- Common source



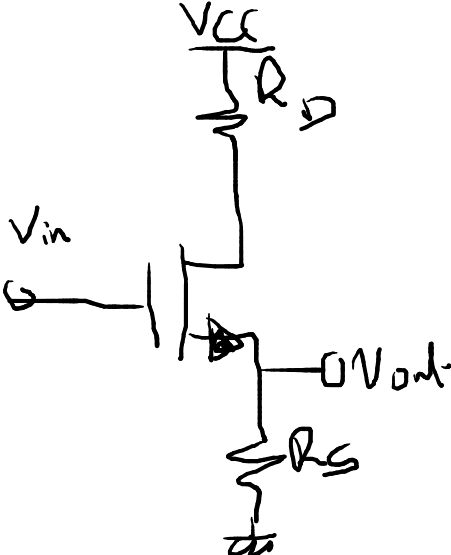
used as a volt and curr amplifier

2- Common gate



volt amplifier, current buffer

3- common drain aka source follower



volt buffer, current amplifier